

24) Prove the second associative law from Table 1 by showing that if A, B and C are sets, then $A \cap (B \cap C) = (A \cap B) \cap C$.

Solution:

$$\begin{aligned}
 A \cap (B \cap C) &= \{x \mid x \in A \wedge x \in (B \cap C)\} \quad (\text{Defn. 2}) \\
 &= \{x \mid x \in A \wedge (x \in B \wedge x \in C)\} \quad (\text{Defn. 2}) \text{ Associative} \\
 &= \{x \mid (x \in A \wedge x \in B) \wedge x \in C\} \quad (\text{2nd } \text{Commutative} \text{ Law of Propositions}) \\
 &= \{x \mid x \in (A \cap B) \wedge x \in C\} \quad (\text{Defn. 2}) \\
 &= (A \cap B) \cap C \quad (\text{Defn. 2}) \quad \square
 \end{aligned}$$

25) Prove the first distributive law from Table 1 by showing that if A, B and C are sets, then $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$.

Solution:

$$\begin{aligned}
 A \cup (B \cap C) &= \{x \mid x \in A \vee x \in (B \cap C)\} \quad (\text{Defn. 1}) \\
 &= \{x \mid x \in A \vee (x \in B \wedge x \in C)\} \quad (\text{Defn. 2}) \\
 &= \{x \mid (x \in A \vee x \in B) \wedge (x \in A \vee x \in C)\} \quad (\text{1st Distributive Law of Propositions}) \\
 &= \{x \mid (x \in A \cup B) \wedge (x \in A \cup C)\} \quad (\text{Defn. 1}) \\
 &= (A \cup B) \cap (A \cup C) \quad (\text{Defn. 2}) \quad \square
 \end{aligned}$$